Experiment 8

AIM: Develop Inventory Management Module using ADO.NET & CRUD Operations

Model file : Product.cs

Code:

```
using System.ComponentModel.DataAnnotations;

namespace ProductModuleInventory.Models
{
    public class Product
    {
        public int ProductId { get; set; }

        [Required]
        public string ProductName { get; set; }

        [Required]
        public int Quantity { get; set; }

        [Required]
        public decimal Price { get; set; }
    }

}
```

Controller file : ProductController.cs

Code:

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Web.Mvc;
using ProductModuleInventory.Models;

namespace ProductModuleInventory.Controllers
{
    public class ProductController : Controller
    {

        string cs = @"Data Source=(localdb)\MSSQLLocalDB;Initial
         Catalog=inventerydb;Integrated Security=True";


        [HttpGet]
        public ActionResult Index()
        {
            DataTable dt = new DataTable();
```

```
    using (SqlConnection con = new SqlConnection(cs))
    {
        string query = "SELECT * FROM Products ORDER BY ProductId";

        using (SqlDataAdapter da = new SqlDataAdapter(query, con))
        {
            da.Fill(dt);
        }
    }

    return View(dt);
}


[HttpGet]
public ActionResult Create()
{
    return View();
}

[HttpPost]
[ValidateAntiForgeryToken]
public ActionResult Create(Product product)
{
    if (ModelState.IsValid)
    {
        using (SqlConnection con = new SqlConnection(cs))
        {
            string query = @"INSERT INTO Products
                (ProductName, Quantity, Price)
                VALUES (@Name, @Quantity, @Price)";

            using (SqlCommand cmd = new SqlCommand(query, con))
            {
                cmd.Parameters.Add("@Name", SqlDbType.VarChar, 100)
                    .Value = product.ProductName;

                cmd.Parameters.Add("@Quantity", SqlDbType.Int)
                    .Value = product.Quantity;

                SqlParameter price = cmd.Parameters.Add(
                    "@Price", SqlDbType.Decimal);
                price.Precision = 10;
                price.Scale = 2;
                price.Value = product.Price;

                con.Open();
                cmd.ExecuteNonQuery();
            }
        }

        return RedirectToAction("Index");
    }
```

```
    return View(product);
}


[HttpGet]
public ActionResult Edit(int id)
{
    Product product = null;

    using (SqlConnection con = new SqlConnection(cs))
    {
        string query =
             "SELECT * FROM Products WHERE ProductId = @Id";

        using (SqlCommand cmd = new SqlCommand(query, con))
        {
            cmd.Parameters.Add("@Id", SqlDbType.Int).Value = id;

            con.Open();

            using (SqlDataReader dr = cmd.ExecuteReader())
            {
                if (dr.Read())
                {
                    product = new Product
                    {
                        ProductId = Convert.ToInt32(dr["ProductId"]),
                        ProductName = dr["ProductName"].ToString(),
                        Quantity = Convert.ToInt32(dr["Quantity"]),
                        Price = Convert.ToDecimal(dr["Price"])
                    };
                }
            }
        }
    }

    if (product == null)
        return HttpNotFound();

    return View(product);
}


[HttpPost]
[ValidateAntiForgeryToken]
public ActionResult Edit(Product product)
{
    if (ModelState.IsValid)
    {
        using (SqlConnection con = new SqlConnection(cs))
        {
            string query = @"UPDATE Products SET
                ProductName = @Name,
                Quantity = @Quantity,
```

```
                    Price = @Price
                    WHERE ProductId = @Id";

                using (SqlCommand cmd = new SqlCommand(query, con))
                {
                    cmd.Parameters.Add("@Name", SqlDbType.VarChar, 100)
                        .Value = product.ProductName;

                    cmd.Parameters.Add("@Quantity", SqlDbType.Int)
                        .Value = product.Quantity;

                    cmd.Parameters.Add("@Id", SqlDbType.Int)
                        .Value = product.ProductId;

                    SqlParameter price = cmd.Parameters.Add(
                        "@Price", SqlDbType.Decimal);
                    price.Precision = 10;
                    price.Scale = 2;
                    price.Value = product.Price;

                    con.Open();

                    int rows = cmd.ExecuteNonQuery();

                    if (rows == 0)
                        return HttpNotFound();
                }
            }

            return RedirectToAction("Index");
        }

        return View(product);
    }

    [HttpPost]
    [ValidateAntiForgeryToken]
    public ActionResult Delete(int id)
    {
        using (SqlConnection con = new SqlConnection(cs))
        {
            string query =
                "DELETE FROM Products WHERE ProductId = @Id";

            using (SqlCommand cmd = new SqlCommand(query, con))
            {
                cmd.Parameters.Add("@Id", SqlDbType.Int).Value = id;

                con.Open();
                cmd.ExecuteNonQuery();
            }
        }

        return RedirectToAction("Index");
```

```
        }
    }

}
```

Content file : Product.css

Code:

```
body {
    font-family: Arial, sans-serif;
    background-color: #f5f5f5;
    color: #333;
}

.inventory-page {
    max-width: 1000px;
    margin: 30px auto;
    padding: 15px;
}

.page-header {
    margin-bottom: 20px;
}

    .page-header h2 {
        color: #234567;
    }

    .page-header p {
        color: #666;
    }

.inventory-card {
    background-color: white;
    padding: 20px;
    border: 1px solid #ddd;
    border-radius: 6px;
}

.inventory-top {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 15px;
}

.product-table {
    width: 100%;
    border-collapse: collapse;
}

    .product-table th {
        background-color: #e8eef5;
        color: #234567;
```

```
    }


    .product-table  th,
    .product-table td {
        padding:  12px;
        border: 1px solid #ddd;
        text-align: left;
    }

    .product-table tr:hover {
        background-color:  #f9f9f9;
    }

.empty-message  {
    text-align: center;
    color: #777;
}

.btn {
    display: inline-block;
    padding: 7px 12px;
    border: none;
    border-radius: 4px;
    text-decoration: none;
    cursor: pointer;
    margin: 2px;
}

.btn-add  {
    background-color:  #2878c7;
    color: white;
}

.btn-edit {
    background-color:  #f0c14b;
    color:  #333;
}

.btn-delete  {
    background-color:  #d9534f;
    color: white;
}

.action-buttons {
    display: flex;
    align-items: center;
    gap: 5px;
}

    .action-buttons form {
        margin: 0;
    }
```

```
.container {
    max-width: 600px;
    margin: 30px auto;
    padding: 20px;
    background-color: white;


    border: 1px solid #ddd;
    border-radius: 6px;
}
.inventory-footer{
    margin-top : 10px;
}
```

View file : Index.cshtml

Code:

```
@using System.Data
@model DataTable

@{
    ViewBag.Title = "Product Inventory";
}

<link href="@Url.Content("~/Content/Product.css")" rel="stylesheet" />

<div class="inventory-page">

    <div class="page-header">
        <h2>Product Inventory</h2>
        <p>Manage your products, stock quantities and prices.</p>
    </div>

    <div class="inventory-card">

        <div class="inventory-top">
            <h3>All Products</h3>

            @Html.ActionLink(
                "+ Add Product",
                "Create",
                "Product",
                null,
                new { @class = "btn btn-add" }
            )
        </div>

        <div class="table-responsive">
            <table class="product-table">
                <thead>
                    <tr>
                        <th>ID</th>
```

```
            <th>Product Name</th>
            <th>Quantity</th>
            <th>Price</th>
            <th>Actions</th>
        </tr>
    </thead>

    <tbody>
        @if (Model == null || Model.Rows.Count == 0)
        {
            <tr>
                <td colspan="5" class="empty-message">
                    No products found. Add your first product!
                </td>
            </tr>
        }
        else
        {
            foreach (DataRow row in Model.Rows)
            {
                <tr>
                    <td>
                        @row["ProductId"]
                    </td>

                    <td class="product-name">
                        @row["ProductName"]
                    </td>

                    <td>
                        <span class="quantity-badge">
                            @row["Quantity"]
                        </span>
                    </td>

                    <td class="price">
                        ₹@Convert.ToDecimal(row["Price"]).ToString("0.00")
                    </td>

                    <td>
                        <div class="action-buttons">

                            @Html.ActionLink(
                                "Edit",
                                "Edit",
                                new { id = row["ProductId"] },
                                new { @class = "btn btn-edit" }
                            )

                            @using (Html.BeginForm(
                                "Delete",
                                "Product",
                                new { id = row["ProductId"] },
                                FormMethod.Post))
```

```
{
    @Html.AntiForgeryToken()
       <button type="submit"
        class="btn btn-delete"
        onclick="return confirm('Are you sure you want
        to delete this product?');">
     Delete  </button>
    }

       </div>
   </td>
</tr>
 }
}
    </tbody>
   </table>
</div>

   <div class="inventory-footer">
     Inventory Management System
   </div>
</div>
</div>
```

View file : Create.cshtml

Code:

```
@model ProductModuleInventory.Models.Product

@{
    ViewBag.Title = "Add Product";
}

<link href="@Url.Content("~/Content/Product.css")" rel="stylesheet" />

<div class="container">
    <h2>Add New Product</h2>
    <p>Enter the details below to add a product to your inventory.</p>

    @using (Html.BeginForm("Create", "Product", FormMethod.Post))
    {
        @Html.AntiForgeryToken()

        <div class="form-group">
            @Html.LabelFor(m => m.ProductName, "Product Name")
            @Html.TextBoxFor(m => m.ProductName,
                new { @class = "form-control", placeholder = "Enter product name" })
            @Html.ValidationMessageFor(m => m.ProductName)
        </div>

        <div class="form-group">
            @Html.LabelFor(m => m.Quantity, "Quantity")
            @Html.TextBoxFor(m => m.Quantity,
```

```
                    new { @class = "form-control", type = "number", min = "0", placeholder =
"Enter quantity" })
                @Html.ValidationMessageFor(m => m.Quantity)
            </div>

            <div class="form-group">
                @Html.LabelFor(m => m.Price, "Price (₹)")
                @Html.TextBoxFor(m => m.Price,
                    new { @class = "form-control", type = "number", step = "0.01", min = "0",
placeholder = "Enter price" })
                @Html.ValidationMessageFor(m => m.Price)
            </div>

            <button type="submit" class="btn btn-success">Save Product</button>
            @Html.ActionLink("Cancel", "Index", "Product",
                null, new { @class = "btn btn-secondary" })
        }
</div>
```

View file : Edit.cshtml

Code:

```
@model ProductModuleInventory.Models.Product


@{
    ViewBag.Title = "Edit Product";
}

<link href="@Url.Content("~/Content/Product.css")" rel="stylesheet" />

<div class="container">
    <h2>Edit Product</h2>
    <p>Update the product information and save your changes.</p>

    @using (Html.BeginForm("Edit", "Product", FormMethod.Post))
    {
        @Html.AntiForgeryToken()
        @Html.HiddenFor(m => m.ProductId)

        <div class="form-group">
            @Html.LabelFor(m => m.ProductName, "Product Name")
            @Html.TextBoxFor(m => m.ProductName,
                new { @class = "form-control", placeholder = "Enter product name" })
            @Html.ValidationMessageFor(m => m.ProductName)
        </div>

        <div class="form-group">

            @Html.LabelFor(m => m.Quantity, "Quantity")
            @Html.TextBoxFor(m => m.Quantity,
                new { @class = "form-control", type = "number", min = "0", placeholder =
```

```
"Enter quantity" })
            @Html.ValidationMessageFor(m => m.Quantity)
        </div>

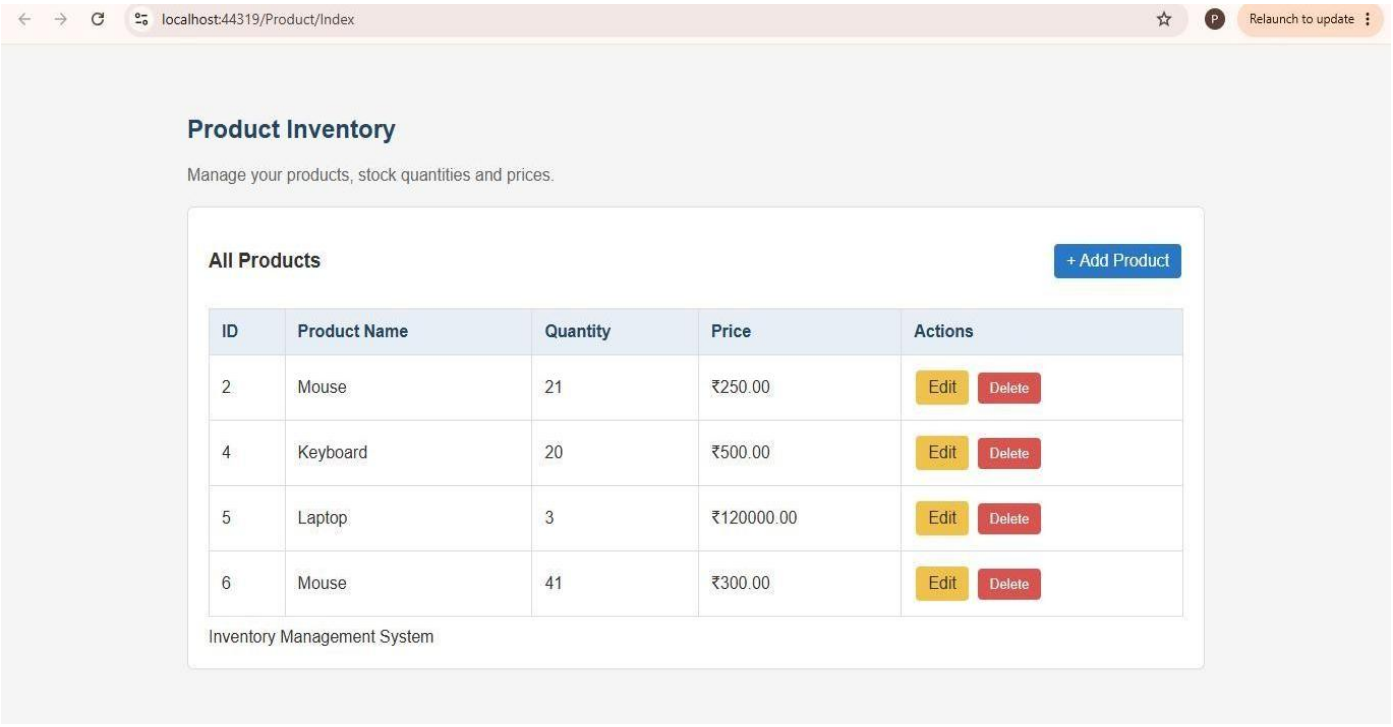
        <div class="form-group">
            @Html.LabelFor(m => m.Price, "Price (₹)")
            @Html.TextBoxFor(m => m.Price,
                new { @class = "form-control", type = "number", step = "0.01", min = "0",
placeholder = "Enter price" })
            @Html.ValidationMessageFor(m => m.Price)
        </div>

        <button type="submit" class="btn btn-success">Update Product</button>

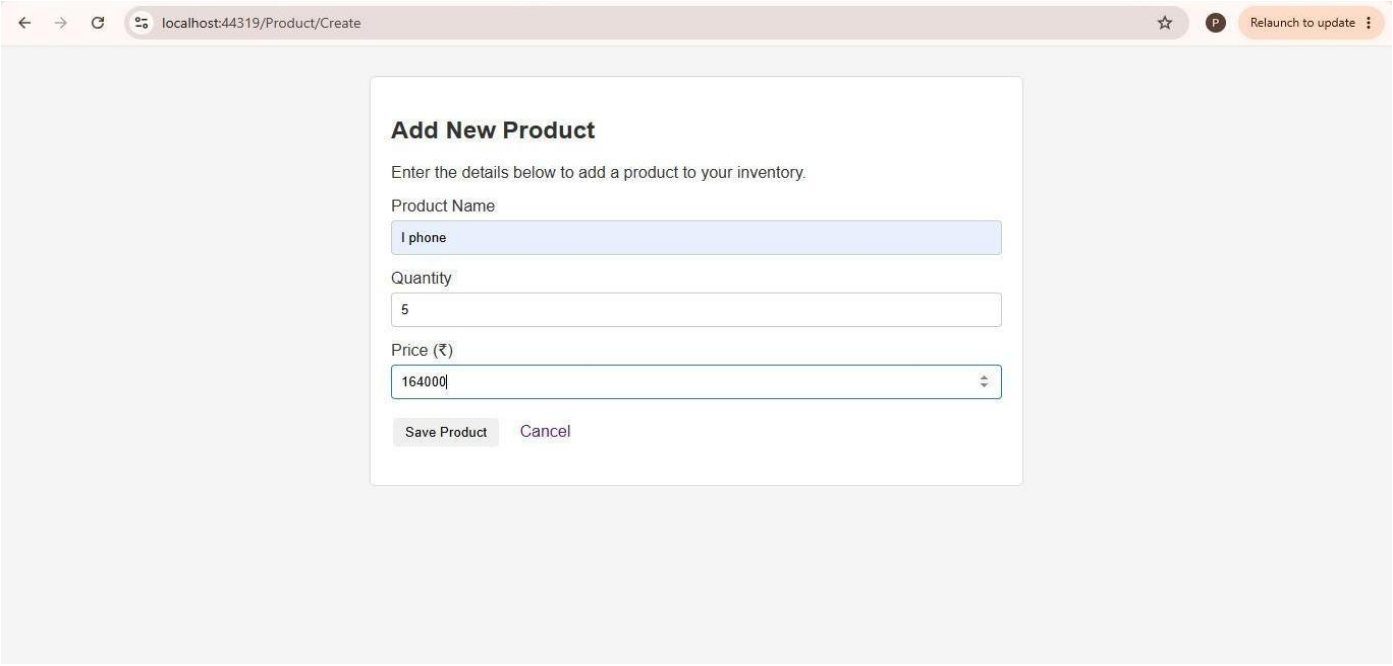
        @Html.ActionLink("Cancel", "Index", "Product",
            null, new { @class = "btn btn-secondary" })
    }

</div>
```

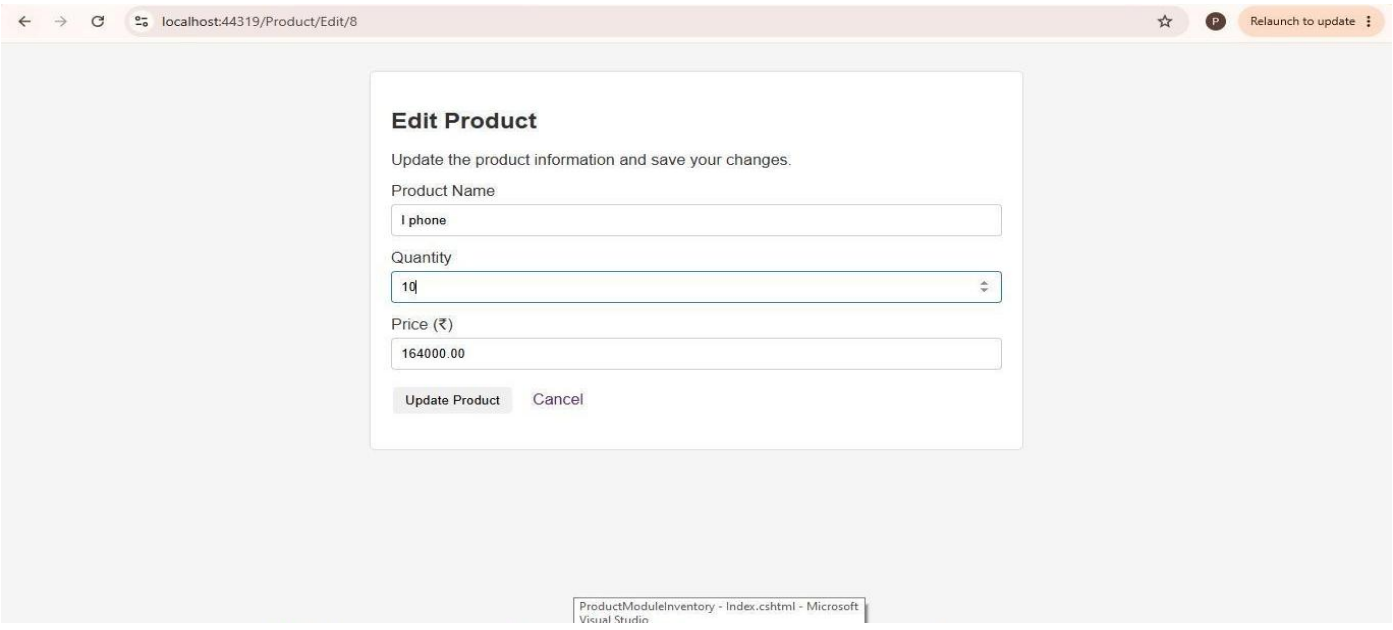
Output:



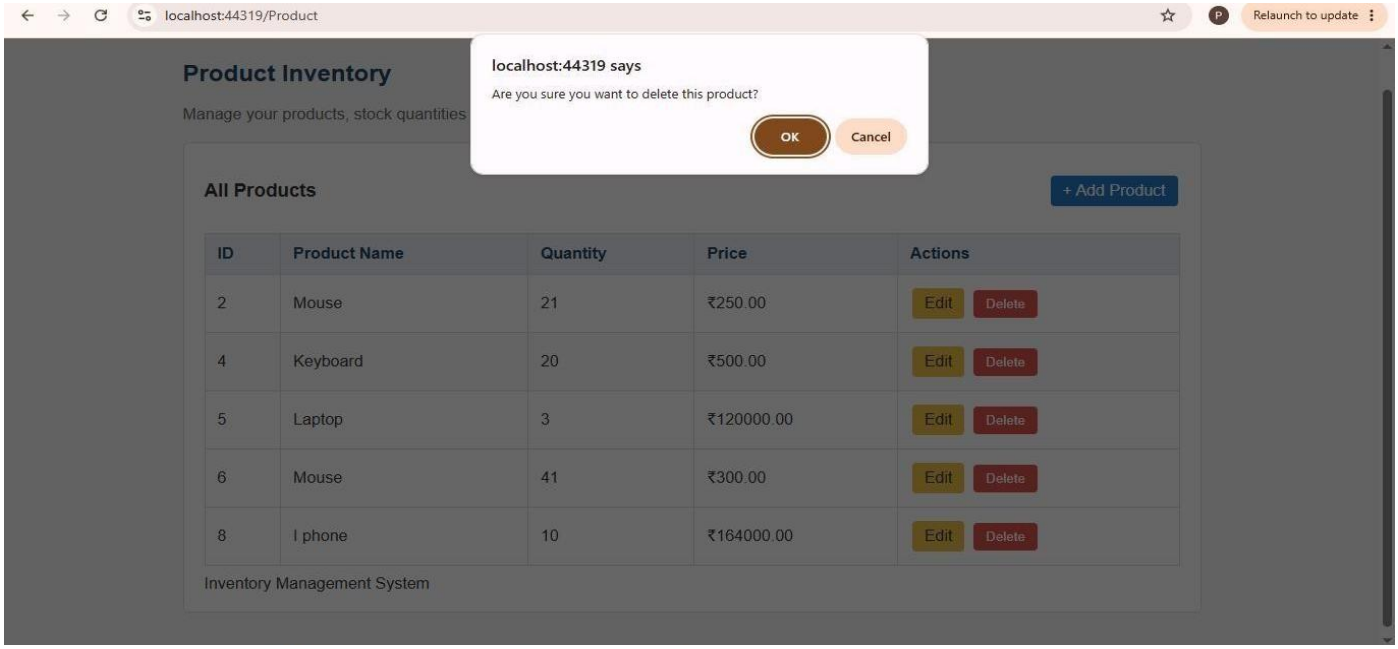
Add Product :



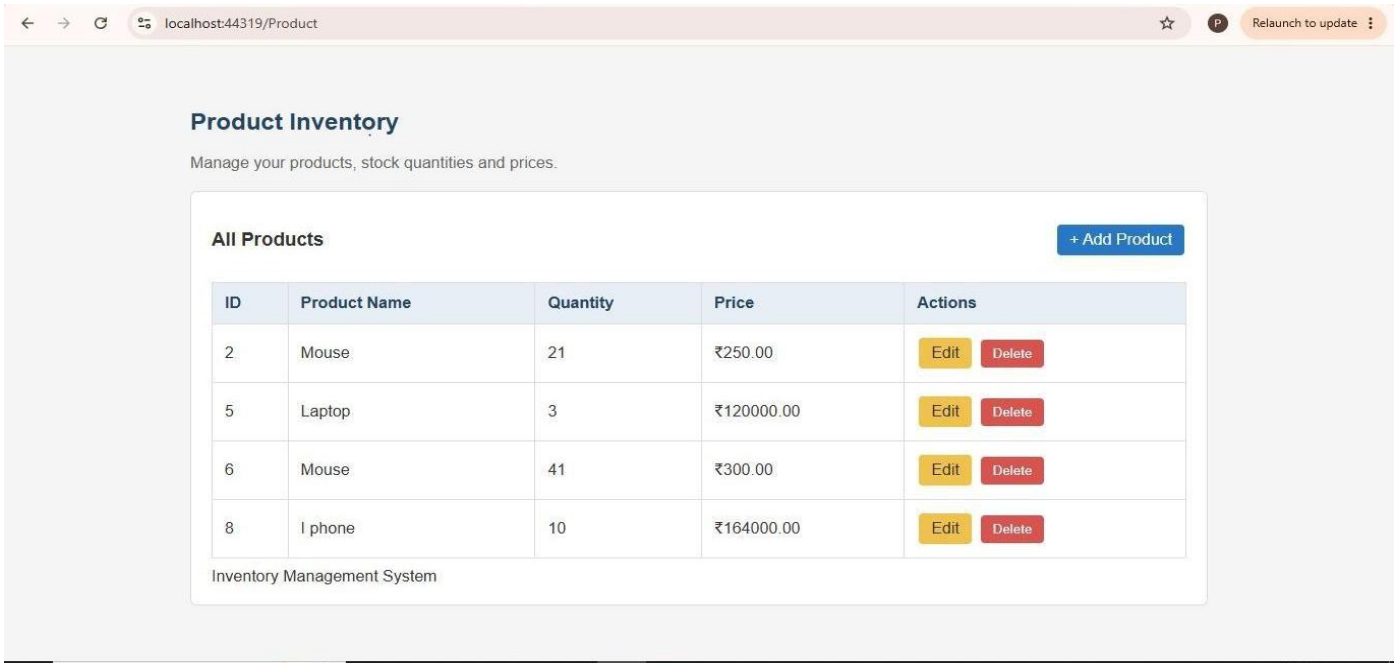
Edit Quantity :



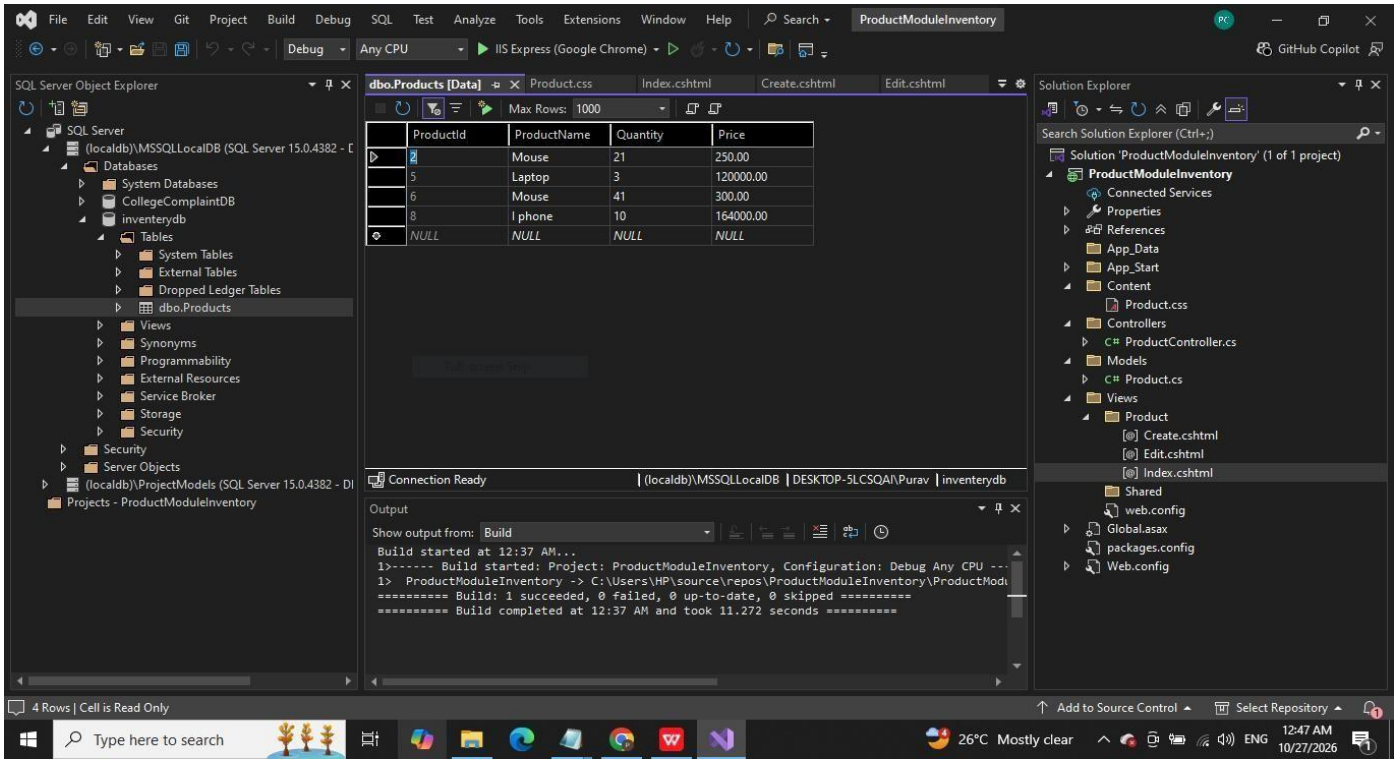
Delete Product :



Update record :



In Database Table Record :



Github URL: https://github.com/bhagyashree1413/Inventory-Management-.git